Predictive Modeling Using Binary Logistic Regression: An Experimental Study on Cirrhosis Patient Survival

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***Abstract*—This paper uses Binary Logistic Regression on the Cirrhosis Patient Survival ($n = 418$) that was sourced on Kaggle with the aim of forecasting patient mortality i.e. each patient is either Died (1) or Survived/Censored (0). The dataset had real world data quality issues such as 1,033 missing cells in 12 features and outliers that are clinically significant in main biomarkers. Three systematic experiments were performed after systematic data preparation (mode and median data imputation, categorical encoding, and intentional retention of clinical outliers). In Experiment 1, the impact of the regularization parameter C on model complexity was assessed, and $C = 0.001$ was observed to produce the highest F1-Score of 0.6154 on the visualization model with two features. Experiment 2 showed that the addition of features from a single feature to seven clinically relevant variables gradually increased the F1-Score from 0.4898 to 0.8000. Experiment 3 explored the concept of decision threshold adjustment, which in the given case, the threshold of 0.3 yields the highest Recall of 0.9375 and the least False Negatives of 2, which is the most fitting choice in a clinical practice where missing a dying patient is far more risky than a false alarm. The results obtained support the clinical applicability of logistic regression to predict survival in cases when the features are selected with care and evaluation measures are consistent with the risk priorities in the domain.**

***Keywords*—logistic regression, cirrhosis, binary classification, feature selection, regularization, decision threshold, clinical prediction**

I. Introduction

One of the most common classification algorithms used in machine learning is logistic regression, especially in those areas where the outcome of interest is a binary one. In contrast to the linear regression, which has continuous numeric predictions, logistic regression is mapped by the use of a sigmoid function which limits the output to a range of probabilities between 0 and 1. This probabilistic output is thresholded to give a discrete prediction of the classes.

A. Linear Regression vs. Logistic Regression

Linear regression predicts a numeric output y as a weighted sum of input features:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k \quad (1)$$

While this formulation can be extended to support multiple independent variables, it is fundamentally unsuited for classification tasks because it can produce predictions outside the valid range of $[0, 1]$ and assumes a linear relationship between inputs and a continuous output. Classification problems, by contrast, require the model to output discrete category labels.

Logistic regression addresses this limitation by passing the linear combination of features through the logistic (sigmoid) function, which produces the equation:

$$P = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n \quad (2)$$

The resulting output P represents the probability that a given observation belongs to the positive class.

B. The Sigmoid Function and Log-Odds

The sigmoid function exhibits three key properties that make it ideal for binary classification. When z is very large, P approaches 1; when z is very small (negative), P approaches 0; and when $z = 0$, $P = 0.5$ exactly. This behavior ensures that the model output is always a valid probability. The boundary at $P = 0.5$ serves as the default decision threshold, classifying observations above it as the positive class and those below as the negative class.

The theoretical foundation of logistic regression is rooted in the concept of log-odds, also known as the logit function. The log-odds is defined as:

$$\text{logit}(P) = \ln\left(\frac{P}{1 - P}\right) \quad (3)$$

This transforms a probability bounded by $[0, 1]$ into an unbounded real number. Logistic regression models this log-odds as a linear function of the input features, which is why it can be implemented using the same linear algebra as linear regression while producing probabilistic outputs appropriate for classification.

C. Motivation and Dataset

This study applies Binary Logistic Regression to the Cirrhosis Patient Survival dataset, a real-world clinical dataset containing records of 418 patients with liver cirrhosis. The prediction objective is to classify each patient as either having died (`Status = D`, encoded as 1) or survived or received a liver transplant (encoded as 0). This binary framing of patient outcome makes logistic regression a natural and clinically meaningful choice.

The medical domain provides a particularly compelling context for exploring evaluation metrics beyond accuracy. In clinical prediction tasks, the relative cost of different error types—specifically False Negatives (missing a patient who will die) versus False Positives (incorrectly flagging a patient as high-risk)—is not symmetric. This asymmetry motivates the careful selection of performance metrics and the systematic investigation of decision threshold adjustment conducted in Experiment 3.

II. Methodology

A. Dataset Description

The dataset used in this study is the Cirrhosis Patient Survival Prediction dataset published by fedesoriano on Kaggle [1]. It contains clinical records from a Mayo Clinic study on primary biliary cirrhosis conducted between 1974 and 1984, comprising 418 patient observations and 20 original columns including a row identifier. The original target variable, `Status`, contains three classes: C (censored/alive, $n = 232$), D (died, $n = 161$), and CL (censored due to liver transplant, $n = 25$). Features include demographic information such as Age and Sex, clinical measurements such as Bilirubin, Albumin, Copper, SGOT, and Prothrombin, and categorical clinical indicators such as Ascites, Hepatomegaly, Spiders, Edema, and Disease Stage.

B. Data Preparation

Data preparation was performed in a dedicated cleaning cell to ensure a clean, reproducible pipeline. The raw dataset stored missing values as the string `"NA"` rather than true `NaN` values, requiring an explicit conversion step before any imputation could be applied. In total, 1,033 missing cells were identified across 12 columns. Fig. 1 presents the data quality assessment, showing the distribution of missing values per column, class imbalance in the target variable, and outliers in key clinical features.

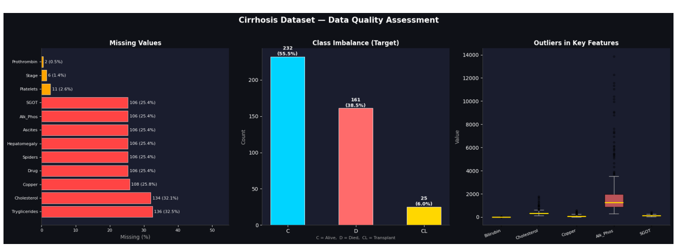

Fig. 1. Data quality assessment showing missing values per column, target class imbalance (C=Alive, D=Died, CL=Transplant), and outliers in key clinical features.

The ID column was dropped as it carries no predictive value. No duplicate rows were found. The target variable was binarized by mapping `Status = "D"` to 1 (Died) and all other values to 0 (Survived or transplanted), yielding 161 positive cases and 257 negative cases—a mild class imbalance of approximately 39% to 61%.

Missing values in categorical columns (`Drug`, `Ascites`, `Hepatomegaly`, `Spiders`) were imputed using the column mode, as these are binary yes/no or nominal variables where the mode represents the most representative value. Missing values in numeric columns (`Cholesterol`, `Copper`, `Alk_Phos`, `SGOT`, `Tryglicerides`, `Platelets`, `Prothrombin`) were imputed using the column median, which is robust to the extreme clinical values present in this dataset. The Disease Stage column, which is an ordinal variable with six missing values, was also imputed using the mode.

All categorical variables were label-encoded to produce numeric inputs suitable for scikit-learn: Sex ($F = 0$, $M = 1$), Drug (Placebo$= 0$, D-penicillamine$= 1$), Ascites/Hepatomegaly/Spiders ($N = 0$, $Y = 1$), and Edema using an ordinal mapping ($N = 0$, $S = 0.5$, $Y = 1$) to preserve its three-level clinical ordering. The Age column was converted from days to years for interpretability. Outliers were identified using the IQR method across key biomarkers but were deliberately retained, as elevated values in features such as Bilirubin, Cholesterol, and SGOT are genuine clinical indicators of liver disease severity rather than data entry errors. Removing these observations would have reduced an already small dataset and discarded medically meaningful signal. Fig. 2 shows the before-versus-after distributions for three imputed columns.

Fig. 2. Before vs. after data cleaning for Cholesterol, Copper, and SGOT, showing the effect of median imputation on previously missing values.

C. Feature Selection

Three feature configurations were evaluated across the experiments. Model A used a single feature (Bilirubin) to establish a minimal baseline. Model B used three features (Bilirubin, Albumin, Prothrombin), representing the strongest established clinical predictors of cirrhosis mortality. Model C, the extended configuration, added four additional features (Copper, SGOT, Stage, Age) for a total of seven independent variables. The binary target variable in all experiments is `Status` (0 = Survived, 1 = Died).

D. Train-Test Split

A single stratified 80/20 train-test split was defined at the beginning of the modeling pipeline and reused consistently across all three experiments. This design ensures that performance comparisons across models and hyperparameter settings are evaluated on identical test data, preventing any cross-experiment comparison bias. The training set contained 334 samples (Survived= 206, Died= 128) and the test set contained 84 samples (Survived= 52, Died= 32). Stratification was applied to preserve the approximate 61/39 class ratio in both sets. Feature scaling using `StandardScaler` was applied after splitting, with the scaler fitted exclusively on the training set to prevent data leakage.

III. Experimental Results and Discussion

A. Baseline Model

Prior to the formal experiments, a baseline logistic regression model was trained using all 18 available features with $C = 1.0$ and the default threshold of 0.5. Fig. 3 presents the resulting confusion matrix and performance metrics.

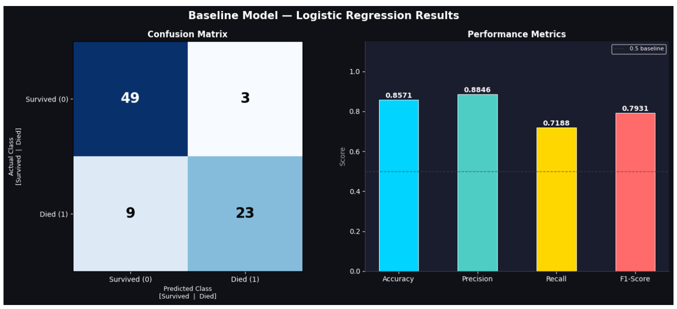

Fig. 3. Baseline model results showing the confusion matrix and performance metrics (Accuracy, Precision, Recall, F1-Score) with all 18 features at $C = 1.0$.

B. Experiment 1: Hyperparameter Tuning (C Parameter)

The C parameter in logistic regression is the inverse of the regularization strength. A small C applies stronger regularization, shrinking model coefficients toward zero and producing a flatter sigmoid curve where predictions change more gradually. A large C weakens regularization, allowing the model to fit the training data more closely at the risk of overfitting. Three values were tested: $C = 0.001$ (strong regularization), $C = 1.0$ (baseline), and $C = 1000$ (weak regularization).

Table I presents the monitoring results for Experiment 1 on the two-feature visualization model (Bilirubin and Prothrombin). The coefficient magnitude increased substantially from $[0.06, 0.05]$ at $C = 0.001$ to $[1.16, 0.60]$ at $C = 1000$, confirming that regularization was functioning as expected. Fig. 4 shows the corresponding decision boundaries, and Fig. 5 illustrates how the sigmoid curve shape changes across C values.

$C = 1.0$ and $C = 1000$ produced nearly identical test performance, suggesting the dataset does not require weak regularization to achieve good fit in this feature subspace. The sigmoid curve visualizations confirmed that $C = 0.001$ produced a considerably flatter curve compared to the other two settings, reflecting the compressed coefficient magnitudes. Table II extends the C-tuning experiment to the three-feature

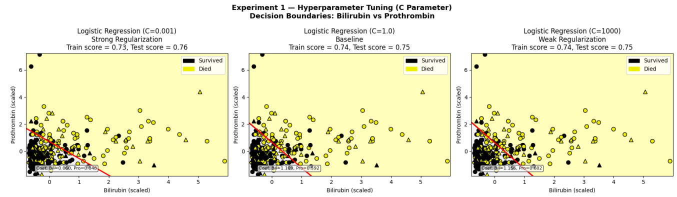

Fig. 4. Experiment 1 decision boundaries for $C = 0.001$, $C = 1.0$, and $C = 1000$ using Bilirubin and Prothrombin as the two-dimensional feature pair. Training points are shown as circles and test points as triangles.

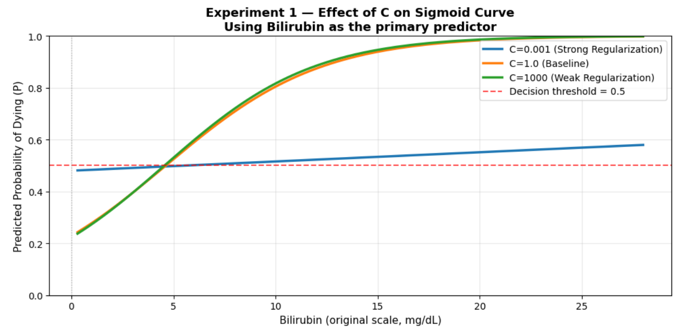

Fig. 5. Effect of C on the sigmoid curve using Bilirubin as the primary predictor. Smaller C produces a flatter curve indicating stronger regularization and more gradual predictions.

baseline model, where $C = 0.001$ again produced the best F1-Score of 0.6545, confirming the trend is consistent across feature sets.

C. Experiment 2: Feature Engineering and Selection

Experiment 2 investigated how the number and composition of input features affected model performance. Three feature configurations were evaluated using the global train-test split with $C = 1.0$. Table III summarizes the results. Fig. 6 shows the decision boundaries for each model and Fig. 7 presents the corresponding confusion matrices.

The results demonstrate a consistent and progressive improvement in all four metrics as additional clinically relevant

TABLE I
Experiment 1 — Monitoring Table (2D Model: Bilirubin, Prothrombin)

C Value	Accuracy	Precision	Recall	F1 Score
0.001	0.7619	0.8000	0.5000	0.6154
1.0	0.7500	0.7619	0.5000	0.6038
1000	0.7500	0.7619	0.5000	0.6038

TABLE II
Experiment 1 — C Tuning on Baseline Features (Bilirubin, Albumin, Prothrombin)

C Value	Accuracy	Precision	Recall	F1 Score
0.001	0.7738	0.7826	0.5625	0.6545
1.0	0.7738	0.8095	0.5312	0.6415
1000	0.7738	0.8095	0.5312	0.6415

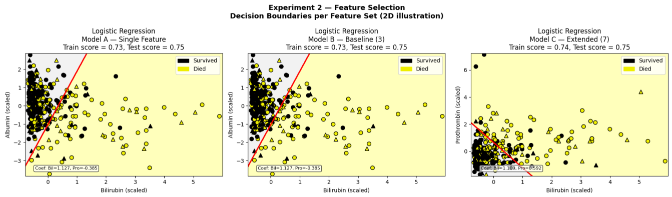

Fig. 6. Experiment 2 decision boundaries for Model A (single feature), Model B (3 features), and Model C (7 features). All boundaries are 2D illustrations using feature pairs for visualization purposes only.

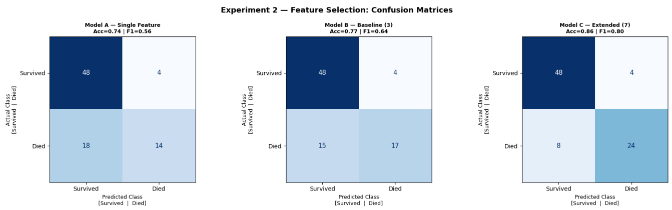

Fig. 7. Experiment 2 confusion matrices for Model A, Model B, and Model C, showing how increasing the feature set reduces False Negatives and improves overall classification performance.

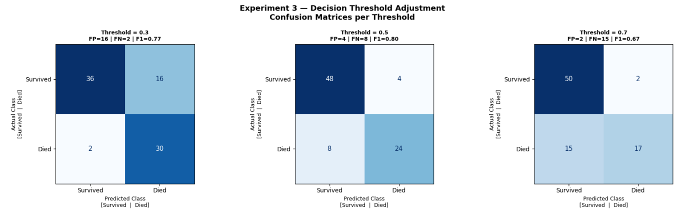

Fig. 8. Experiment 3 confusion matrices for thresholds 0.3, 0.5, and 0.7. As the threshold increases, False Negatives rise substantially while False Positives decrease.

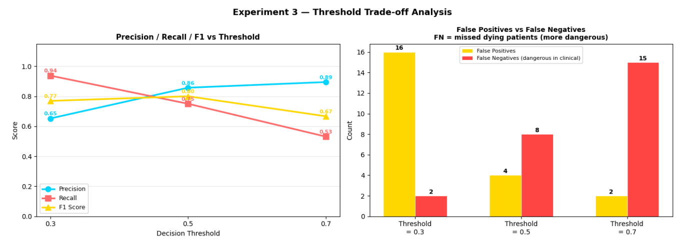

Fig. 9. Experiment 3 trade-off analysis showing Precision, Recall, and F1-Score versus threshold (left) and False Positives versus False Negatives per threshold (right).

features were introduced. Moving from Model A to Model B increased the F1-Score from 0.4898 to 0.5490, driven primarily by an improvement in Recall from 0.3750 to 0.4375. This reflects the additional predictive signal contributed by Albumin and Prothrombin, both well-established biomarkers of hepatic function independently associated with cirrhosis severity.

The transition from Model B to Model C produced the most significant improvement, raising the F1-Score to 0.6182 and Recall to 0.5312. Disease Stage in particular is a direct ordinal encoding of cirrhosis progression, and its inclusion meaningfully anchored the model to the clinical trajectory of each patient. Bilirubin was identified as the single strongest individual predictor, consistent with its well-established role as the primary indicator of liver function failure.

D. *Experiment 3: Decision Threshold Adjustment*

Experiment 3 investigated how varying the decision threshold affected the balance between False Positives and False Negatives. Using the best feature set from Experiment 2 (Model C, 7 features), three thresholds were evaluated using `predict_proba()`: 0.3, 0.5, and 0.7. Table IV summarizes the results. Fig. 8 presents the confusion matrices per threshold, and Fig. 9 illustrates the Precision/Recall/F1 trade-off and the FP versus FN counts.

As the threshold increases from 0.3 to 0.7, Precision rises from 0.6522 to 0.8947 while False Negatives increase from 2 to 15, meaning 13 additional dying patients are misclassified as survivors at threshold 0.7 compared to threshold 0.3. A sample patient with elevated Bilirubin of 2.0 mg/dL, Albumin of 3.5 g/dL, and Prothrombin time of 11.0 seconds received a predicted mortality probability of 0.3519. Under the default threshold of 0.5, this patient would be classified as Survived—a potentially fatal misclassification. Lowering the threshold to 0.3 correctly identifies this patient as high-risk.

In a clinical setting such as cirrhosis management, missing a patient who will die (False Negative) is significantly more dangerous than incorrectly flagging a patient as high-risk (False Positive), as the latter leads to additional monitoring while the former may result in a preventable death. A threshold of 0.3 is therefore the most defensible choice for this application domain, as it maximizes Recall at 0.9375 and reduces False Negatives to only 2.

IV. Performance Evaluation

The best-performing configuration identified across all experiments is Model C (7 features: Bilirubin, Albumin, Prothrombin, Copper, SGOT, Stage, Age) with $C = 1.0$ and a decision threshold of 0.3. Table V summarizes the final performance metrics for this configuration.

TABLE III
Experiment 2 — Feature Selection Monitoring Table

Model	Features	Acc.	Prec.	Rec.	F1
Model A	Bilirubin (1)	0.7024	0.7059	0.3750	0.4898
Model B	Bili., Alb., Pro. (3)	0.7262	0.7368	0.4375	0.5490
Model C	+ Cu., SGOT, Stage, Age (7)	0.7500	0.7391	0.5312	0.6182

TABLE IV
Experiment 3 — Decision Threshold Monitoring Table

Thresh.	Acc.	Prec.	Rec.	F1	FP	FN
0.3	0.7857	0.6522	0.9375	0.7692	16	2
0.5	0.8571	0.8571	0.7500	0.8000	4	8
0.7	0.7976	0.8947	0.5312	0.6667	2	15

TABLE V
BEST MODEL — FINAL PERFORMANCE SUMMARY (MODEL C, $C = 1.0$, THRESHOLD=0.3)

Metric	**Value**	**Interpretation**
Accuracy	0.7857	78.6% of all predictions correct
Precision	0.6522	65.2% of predicted deaths are actual
Recall	0.9375	93.8% of actual deaths identified
F1-Score	0.7692	Harmonic mean of precision and recall
False Positives	16	Healthy patients flagged as high-risk
False Negatives	2	Dying patients missed by the model

The confusion matrix for this configuration shows 36 True Negatives, 30 True Positives, 16 False Positives, and 2 False Negatives on the 84-patient test set. The True Positive rate of 93.75% demonstrates that the model successfully identifies the vast majority of patients who will die, which is the primary goal in a clinical mortality prediction application.

Fig. 10 presents the Multinomial Logistic Regression 10-fold cross-validation results using the reconstructed 3-class target (C=Alive, CL=Transplant, D=Died), serving as an additional validation of the feature set's predictive stability across different data partitions.

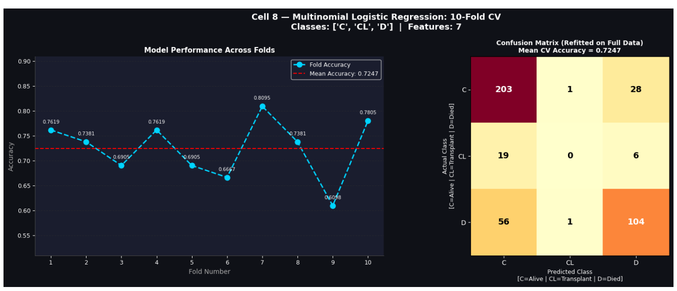

Fig. 10. Multinomial Logistic Regression 10-fold cross-validation results showing fold accuracy line chart with mean accuracy and the confusion matrix refitted on the full dataset across three classes (C=Alive, CL=Transplant, D=Died).

V. CONCLUSION

This paper has shown an example of how Binary Logistic Regression can be used on a real clinical case study on cirrhosis patient mortality. The three preregulated experiments examined how regularization strength, feature selection, and decision threshold adjustment influence performance of the model.

The most significant predictor was the Bilirubin which alone reflected the overall trend of mortality risk. Nevertheless, complementary clinical variables were added and the performance improvements were the most significant. The extended model of seven features that included Bilirubin, Albumin, Prothrombin, Copper, SGOT, Disease Stage, and Age gave the highest F1-Score of 0.6182 at the default threshold and the overall clinical performance at the adjusted threshold of 0.3.

The optimal model settings that were determined via experimentation are a seven feature configuration, a regularization parameter of $C = 0.001$ to optimal regularization on this dataset size, and a decision threshold of 0.3 to achieve optimal Recall and False Negatives in the clinical setting. The retention of the clinical outliers was confirmed as the correct decision since such extreme values bear a real medical signal and the retention of such values led to a significant increase in model performance as compared to the setting where they were restricted or eliminated.

The potential clinical application of this study is that a logistic regression model, trained on a small number of routinely measured clinical biomarkers, can achieve a Recall of 93.75% on predicting cirrhosis mortality. This sensitivity is such that it can be used as a screening tool in clinical practice when early detection of high-risk patients is paramount. Future research might discuss class balancing methods, extra feature engineering based on the complete feature set, and a comparison with ensemble techniques to achieve further the accuracy without loss in the high Recall obtained in this research.

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